Design and Evaluate Different Prompt Types to Control Agent Reasoning, Autonomy, and Output Quality

AIM

To design and evaluate different prompt types used in AI agents and analyze their influence on reasoning ability, autonomy, accuracy, completeness, and overall output quality.

OBJECTIVE

The main objective of this experiment is to understand how different prompting techniques influence the performance of AI agents. The experiment examines how the structure, clarity, and complexity of a prompt affect an agent's ability to understand instructions, solve problems, make decisions, use tools, and generate accurate and meaningful responses.

It also aims to compare different prompt types based on reasoning, autonomy, and output quality and identify suitable prompting techniques for simple tasks and complex agentic workflows.

THEORY

Prompting is the process of providing instructions, context, examples, or constraints to an artificial intelligence model to guide its response. In AI agents, prompts play an important role in determining how the agent interprets a task, selects actions, uses available tools, and produces the final output.Different prompt types provide different levels of guidance. Simple prompts may be sufficient for basic tasks, whereas complex tasks may require detailed instructions, multiple examples, self-evaluation, or toolassisted reasoning. Therefore, selecting an appropriate prompt type helps improve the accuracy, consistency, clarity, and reliability of AI-generated responses.

An AI agent's performance can be evaluated using parameters such as reasoning ability, autonomy, completeness, hallucination rate, clarity, and response time. By testing different prompting techniques under comparable conditions, their strengths and limitations can be identified.

Prompt Type	Description	Example
Zero-shot	Task only, no examples.	Summarize this article in 100 words.
Few-shot	Provide examples before the task.	Example summaries then a new article.
Chain-of-Thought	Encourage step-by-step reasoning.	Solve step by step.
Role-based	Assign an expert role.	You are an experienced software archit
Instruction-based	Detailed instructions and constraints.	Generate Python code with comments.

Self-Reflection	Review and improve own answer.	Review your solution and improve it.
ReAct	Reason while taking actions/tools.	Think, search if needed, then answer.
Tree-of-Thought	Explore multiple solution paths.	Generate three solutions and compare.

TYPES OF PROMPTING TECHNIQUES

4.1 Zero-Shot Prompting

Zero-shot prompting involves providing a task directly to the AI model without supplying any examples. The model generates a response based on its existing learned knowledge and its understanding of the instruction.

For example, the prompt "Summarize this article in 100 words" asks the model to summarize an article without showing examples of previously written summaries.

This technique is simple, requires minimal preparation, and is suitable for straightforward tasks. However, the output may not always follow the expected format or level of detail when the instructions are ambiguous.

4.2 Few-Shot Prompting

Few-shot prompting provides a small number of examples before presenting the actual task. These examples demonstrate the expected format, style, or pattern of the response.

For example, the model may be given two sample article summaries before being asked to summarize a new article in the same style.

This technique helps the model understand the expected output and improves consistency across similar tasks. It is useful when a particular response format is required, although providing examples increases the length of the prompt.

4.3 Chain-of-Thought Prompting

Chain-of-thought prompting encourages the model to approach a problem through intermediate reasoning steps rather than producing only an immediate answer.

For example, a prompt may ask the model to solve a mathematical problem step by step and then provide the final answer.

This approach can help structure complex problem-solving tasks and make explanations easier to follow. However, a detailed explanation does not automatically guarantee correctness, so the final answer must still be evaluated for accuracy.

4.4 Role-Based Prompting

Role-based prompting assigns a specific role or professional perspective to the AI model. This helps establish the context, terminology, and style expected in the response.

For example, the prompt "You are an experienced software architect. Design a scalable web application architecture" directs the model to respond from a software architecture perspective.

This technique is useful for domain-specific tasks, technical explanations, and professional recommendations. Its effectiveness depends on the clarity of the task and the information provided, rather than the assigned role alone.

4.5 Instruction-Based Prompting

Instruction-based prompting provides detailed instructions, requirements, and constraints that the model must follow while generating its response.

For example, "Generate Python code with comments, include input validation, handle exceptions, and explain how the program works" specifies the expected features of the output.

This technique provides strong control over the response format and task requirements. It is especially useful for programming, report generation, data processing, and other tasks in which completeness and adherence to constraints are important.

4.6 Self-Reflection Prompting

Self-reflection prompting asks the model to review its initial response, identify possible errors or omissions, and produce an improved version.

For example, the prompt "Review your solution for logical errors, missing requirements, and unclear explanations, then improve it" encourages an additional evaluation stage.

This technique can improve completeness, clarity, and consistency by encouraging the model to reconsider its initial answer. However, self-review may not identify every error, particularly when the model lacks reliable information or an independent verification method.

ReAct Prompting

ReAct stands for Reasoning and Acting. It combines reasoning with actions, allowing an AI agent to interact with external tools or information sources when required.

For example, an agent may identify the information needed to answer a question, search an available source, examine the retrieved information, and formulate a response based on the results.

This technique is useful for research, information retrieval, and tasks that require multiple actions. It supports greater autonomy because the agent can select and execute appropriate actions within its permitted tools and instructions. The quality of the result depends on the reliability of the tools and the information retrieved.

Tree-of-Thought Prompting

Tree-of-thought prompting encourages the exploration of multiple possible solution paths before selecting a final approach.

For example, a prompt may ask the model to generate three possible solutions to a problem, compare their advantages and limitations, and select a suitable approach according to specified requirements.

This technique is useful for planning, complex decision-making, and problems with several possible solutions. It allows alternatives to be examined systematically, although exploring multiple paths may increase computational effort and response time.

PROCEDURE

1. Identify the task to be performed by the AI agent and define the expected output.
2. Prepare different prompts using zero-shot, few-shot, chain-of-thought, role-based, instruction-based, self-reflection, ReAct, and tree-of-thought techniques.
3. Submit the prompts to the AI agent and obtain the corresponding responses.
4. Examine each response for correctness, completeness, logical reasoning, clarity, and compliance with the given instructions.
5. Evaluate the agent's autonomy by examining its ability to identify suitable actions, consider alternatives, and use available tools when required.
6. Record the performance of each prompting technique using a rating scale of 1 to 5, where 1 indicates low performance and 5 indicates high performance.
7. Compare the recorded results to identify differences in reasoning, autonomy, and output quality.
8. Analyze the results and determine which prompting techniques are suitable for simple tasks and which are more appropriate for complex agentic workflows.

EVALUATION CRITERIA

The performance of each prompting technique is evaluated using the following criteria.

- Accuracy: Measures whether the generated response is factually and logically correct.
- Completeness: Determines whether the response covers all the required information and satisfies the task requirements.
- Reasoning: Evaluates the logical structure of the solution and the quality of the explanation provided.
- Autonomy: Measures the agent's ability to make task-related decisions and select appropriate actions independently within the given instructions and available capabilities.
- Hallucination: Examines whether the model produces unsupported, fabricated, or incorrect information. A lower hallucination rate indicates better reliability.
- Clarity: Measures how clearly and understandably the response is presented.

- Response Time: Measures the time required to generate the response. Lower response time is generally preferable when other performance requirements are satisfied.

RESULT ANALYSIS AND DISCUSSION

Prompt	Reasoning	Autonomy	Output Quality	Overall
Zero-shot	2/5	2/5	3/5	2.3
Few-shot	3/5	3/5	4/5	3.3
Role-based	4/5	3/5	4/5	3.7
Instruction-based	4/5	4/5	5/5	4.3
Chain-of-Thought	5/5	4/5	5/5	4.7
Self-Reflection	5/5	5/5	5/5	5.0
ReAct	5/5	5/5	5/5	5.0
Tree-of-Thought	5/5	5/5	5/5	5.0

The sample results indicate that prompting techniques influence the reasoning, autonomy, and output quality of AI agents.

Zero-shot prompting received comparatively lower scores because it provides only the task instruction without examples or additional guidance. Few-shot prompting showed improved output quality because examples help establish the expected response pattern. Role-based prompting improved reasoning and output quality by providing a specific context for the task.

Instruction-based prompting achieved an overall score of 4.3 because detailed requirements and constraints provide better control over the generated output. Chainof-thought prompting received an overall score of 4.7, indicating strong performance in tasks requiring structured problem-solving and logical explanations.

Self-reflection, ReAct, and tree-of-thought prompting received the highest sample scores of 5.0. Self-reflection supports response revision, ReAct enables interaction with tools and external information, and tree-of-thought prompting explores alternative solution paths. These capabilities can be useful in complex tasks that require verification, planning, or multiple decisions.

However, the scores represent the sample evaluation shown in this experiment and should not be interpreted as universal performance guarantees. Actual results may vary depending on the task, model, prompt design, tools, and evaluation method.

CONCLUSION

Thus, different prompt types were designed and evaluated to understand their influence on AI agent reasoning, autonomy, and output quality. The sample evaluation showed that zero-shot prompting is suitable for simple tasks, while few-shot and rolebased prompting provide additional guidance and context. Instruction-based prompting offers effective control over output requirements, whereas chain-of-thought, selfreflection, ReAct, and tree-of-thought prompting support more structured problemsolving, review, tool interaction, and exploration of alternatives.

The experiment demonstrates that selecting an appropriate prompting technique according to task complexity and requirements can improve the usefulness and reliability of AI agents.